CNV-profile regression for copy number variant association analysis with whole genome sequencing data

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Goal

- **Copy number variants (CNVs)**
  DNA duplications or deletions involving at least 50bp
- We propose CNV-profile regression
  Estimating CNV effects on phenotype
- **Properties of the proposed methods**
  - Incorporate **multiple CNV features** (see Fig 1)
    - Dosage (dup/del, number of extra copies)
    - Length
  - Jointly assess all CNV events within a genomic region
  - Avoid arbitrary CNV region definition
  - Allow CNV effects comparable across different studies

Simulations

Different causal patterns

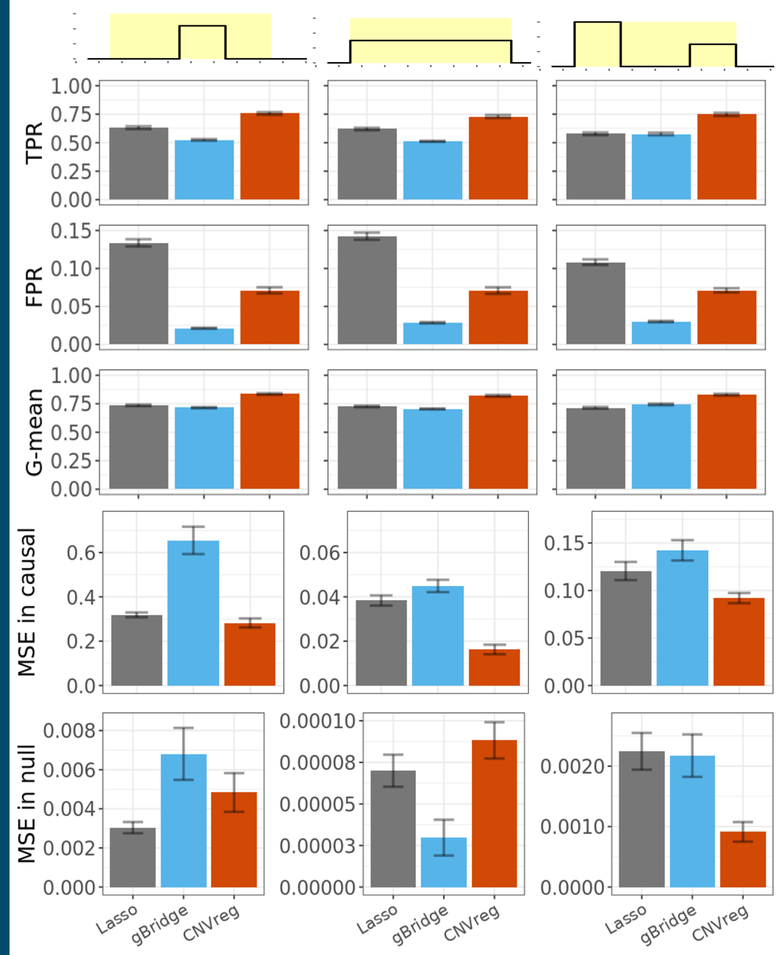

Fig 2. Variable selection (TPR, FPR, and Geometric Mean) and effect estimation (MSE) performance

Methods

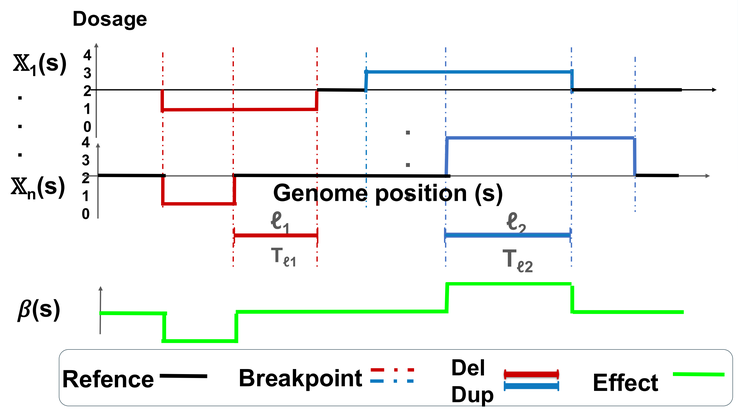

Fig 1. CNV profile curve $X_i(s)$ and CNV effect curve β(s)

Model interpretation:

- Under the $\beta(s) = \beta_l \; for \; s \in frag\, l$ definition, β_l of the same genomic position is comparable across different studies
- In contrast, in a commonly considered model,
  $$g(\mu_i) = \alpha_0 + \sum_l \alpha_l |Dosage_{il} - 2| = \alpha_0 + \sum_l \alpha_l \frac{x_{il}}{T_l}$$
  α_l is the aggregated CNV effect across all positions in a fragment. That is, effect α_l is length-dependent
- Then, $\alpha_l / T_l = \beta_l$ is comparable across studies

$$g(\mu_i) = \beta_0 + \int_s \beta(s) x_i^*(s) ds = \beta_0 + \sum_{frag\, l} \int_{s \in frag\, l} \beta(s) x_i^*(s) ds$$

$$x_i^*(s) \equiv |Dosage_{is} - 2|$$

Dosage of subject i at position s

Refence copy number

| absolute value | accommodates duplication and deletion

- Assume constant effect within a CNV fragment: $\beta(s) = \beta_l \; for \; s \in frag\, l$
- Define "Length" × "Dosage" in design matrix
  $$x_{il} = \int_{s \in frag\, l} x_i(s) ds = T_{il} \times |Dosage_{il} - 2|$$
- The proposed model can be simplified as
  $$g(\mu_i) = \beta_0 + \sum_l \beta_l x_{il}$$
  Regress trait on "Length" × "Dosage"

Model fitting with 2 penalty terms:

- Lasso penalty: to deal with high dimensionalities and to select phenotype-associated CNVs, i.e, $||\beta||_1 < a$
- Weighted fusion: to encourage (not force) adjacent CNV fragments to share similar effect size, i.e., $\sum_l w_{l,l+1} (\beta_{l+1} - \beta_l)^2 < b$

Application to real data

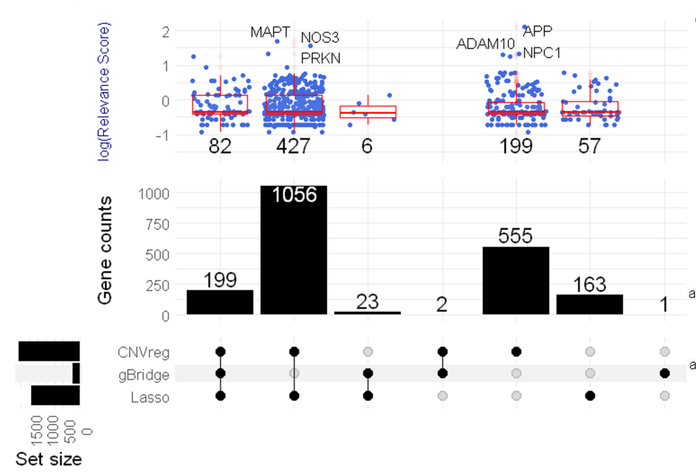

Fig 3. Number of genes interrupted by CNVs identified by different methods and boxplots of non-zero Alzheimer's diseases-relevance scores (log10 scale) from GeneCard.

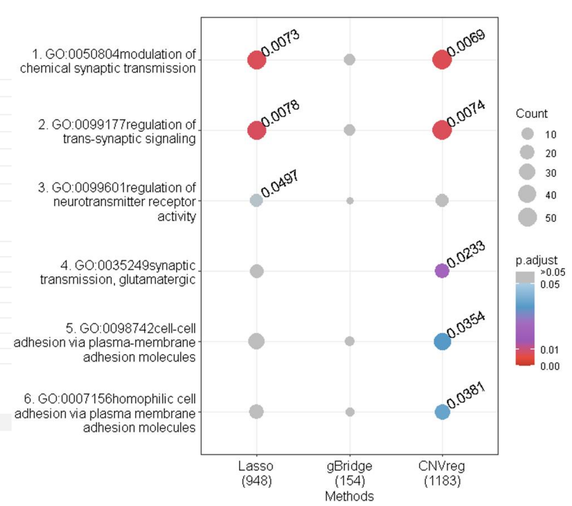

Fig 4. Gene ontology enrichment analysis for Alzheimer's diseases associated genes identified by different methods

Summary

The proposed model:

- Better in variable selection
- Better in effect size estimation
- Identifies additional genes with high AD-relevance score and identifies more enriched genes in AD-related pathways (in neuron structure and function)

Preprint

CNVreg R package

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